

Docker Image Build Failing SOP

Objective

Resolve Docker image build failures caused by missing dependencies, broken package installation, invalid Dockerfile syntax, or unavailable build context files.

Scope

Applicable to Docker-based services, containerized application deployments, DevOps teams, SRE teams, platform teams, and synthetic incident workflows across development, staging, and production-like environments.

Pre-requisites

1. Access to Docker host, Docker CLI, container runtime logs, and the target container or image name.
2. Permission to inspect containers, images, volumes, networks, resource limits, and compose files where applicable.
3. Access to application logs, deployment artifacts, Dockerfile, environment variables, and registry credentials.
4. Monitoring access for host CPU, memory, disk, container restart count, and container health checks.
5. Escalation contact for application, DevOps, security, registry, infrastructure, and platform teams.

Incident Metadata

Field	Value
Ticket ID	T004
Issue	Docker image build failing
Category	Build
Service	Docker
Severity	Medium
Status	Resolved
Description	Docker build fails due to missing dependencies
Expected Resolution	Update Dockerfile and install required packages

Related Log Evidence

Log ID	Timestamp	Level	Message	Error Code
N/A	N/A	N/A	No direct log evidence found in logs CSV for this ticket.	N/A

Procedure

1. Capture the build command, Dockerfile path, build context, image tag, and failed build step.
2. Review the build output and identify the first failing instruction.
3. Validate Dockerfile syntax, base image availability, package manager commands, and copied file paths.
4. Check whether required dependency files such as requirements.txt, package.json, or pyproject.toml exist in the build context.
5. Rebuild with plain progress logs to get complete error details.
6. Update Dockerfile instructions, dependency installation, or build context configuration.
7. Run a clean build and confirm the image is created successfully.
8. Push the image to the registry only after scanning and smoke testing.

Common Commands

```
docker ps -a
docker logs <container-name-or-id> --tail=100
docker inspect <container-name-or-id>
docker stats --no-stream
docker images
docker network ls && docker network inspect <network-name>
docker volume ls && docker volume inspect <volume-name>
docker system df
```

Troubleshooting Matrix

Issue	Possible Cause	Resolution
Build fails	Missing dependency	Install required package in Dockerfile
COPY failed	File not in build context	Fix path or build context
Package install fails	Repository or version issue	Pin compatible package version
Base image pull fails	Registry or image tag issue	Check base image name and access

Best Practices

1. Keep Dockerfiles small and deterministic.
2. Pin dependency versions for repeatable builds.
3. Use .dockerignore to avoid unnecessary context files.
4. Run builds in CI before production deployment.

References

Docker Docs | Internal Tickets CSV | Internal Logs CSV | Container Runtime Logs | Monitoring Dashboards | DevOps Runbook